

EXPERIMENT-3

Aim : Managing Source Code on GitHub using Git.

1. GitHub

GitHub is a cloud-based platform used to **host, manage, and collaborate on Git repositories**. It allows developers to store source code remotely and work together on projects.

2. Remote Repository

A **remote repository** is a repository stored on a remote server such as GitHub. It allows developers to share and synchronize their source code.

3. Local and Remote Repository

- Local Repository ↔ GitHub Remote Repository
Local Repository: Repository available on the developer's computer.
- **Remote Repository:** Repository hosted on GitHub.

4. Managing Source Code with GitHub

Git and GitHub can be used to:

- Upload source code to GitHub.
- Download/clone existing repositories.
- Synchronize local and remote changes.
- Maintain version history.
- Collaborate with other developers.
- Manage different branches.

5. Basic GitHub Workflow

Create/Modify Source Code



Local Repository



GitHub



Remote Repository



Collaboration / Sharing

6. Important Concepts

- **Repository** – Stores project files and their history.
- **Commit** – Records changes in the repository.
- **Branch** – Separate development line.
- **Remote** – Connection to a remote repository.
- **Clone** – Creates a local copy of a remote repository.
- **Push** – Sends local changes to GitHub.
- **Pull** – Gets changes from GitHub.

1. Launch the Git Bash Terminal



2. Configure User Information



3. Create a Project and Initialize Git

`mkdir food-guests`

`cd food-guests`

`git init`

A screenshot of a Windows Command Prompt window with a blue title bar. The title bar text is "MINGW64:/c:/Users/itsw2a/projects/devops/food-guests". The terminal shows three commands being executed: 1. "itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops" followed by "\$ mkdir food-guests". 2. "itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops" followed by "\$ cd food-guests". 3. "itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests" followed by "\$ mkdir main test".

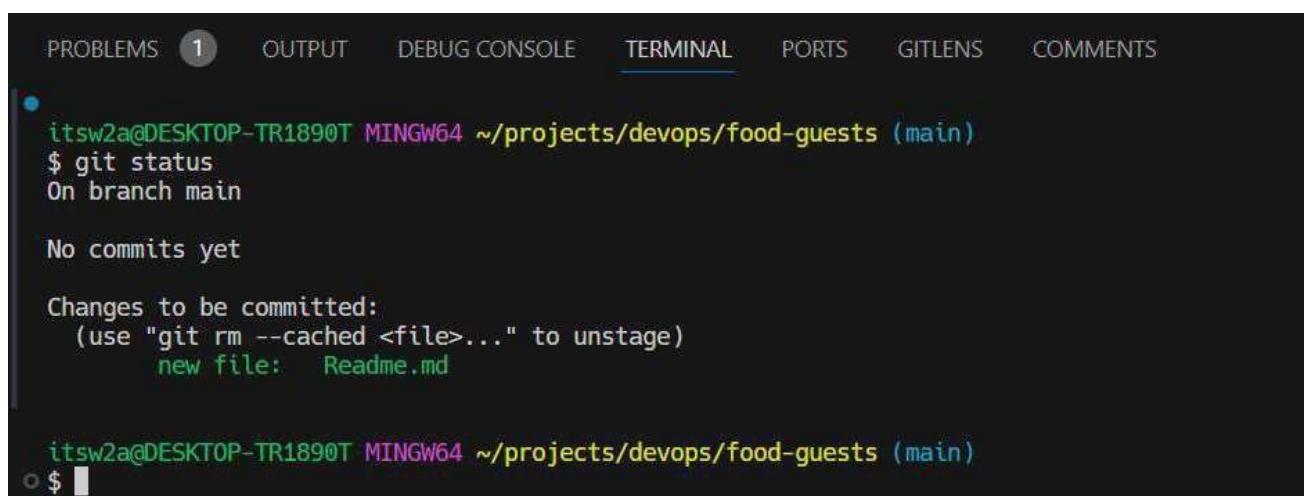
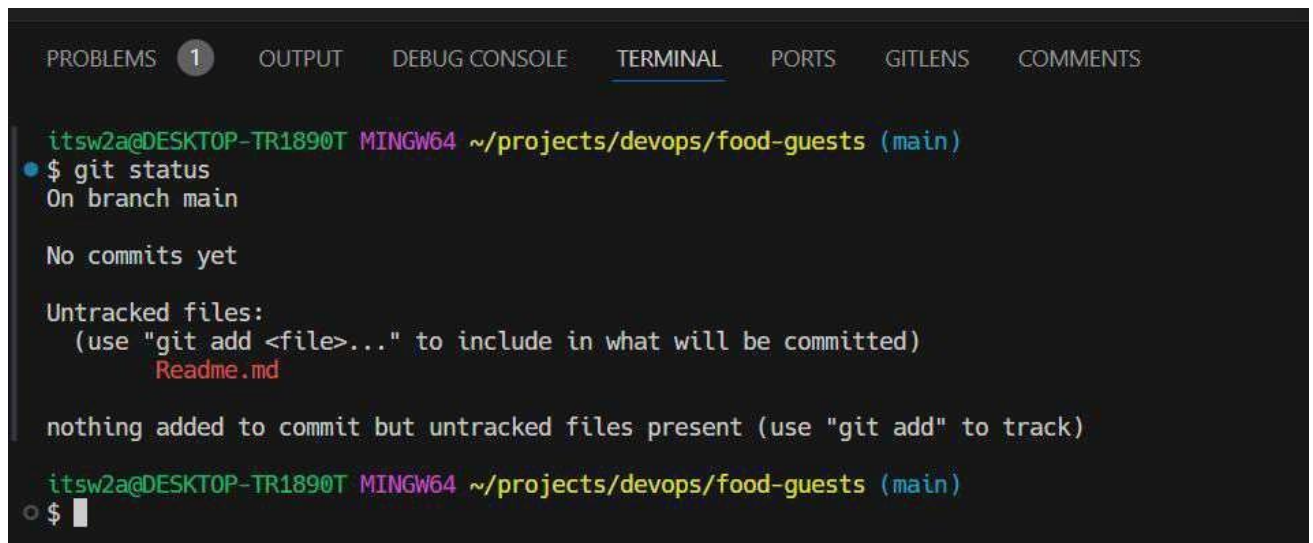
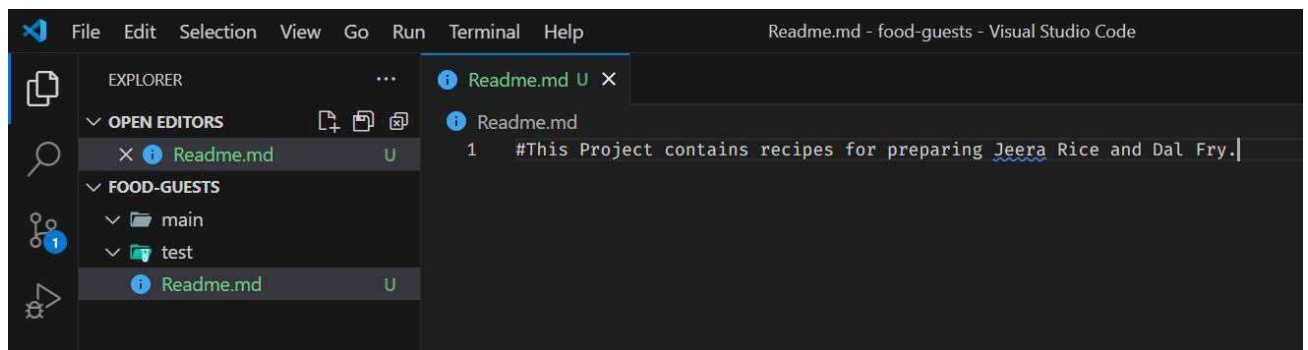
```
MINGW64:/c:/Users/itsw2a/projects/devops/food-guests  
itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops  
$ mkdir food-guests  
itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops  
$ cd food-guests  
itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests  
$ mkdir main test
```

4. Make and Stage the First Change

(Create and edit Readme.md in your editor)

`git status`

`git add Readme.md`



5. Commit the Changes

```
git commit -m "Initial commit"
```

```
git log
```

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS COMMENTS
itstw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
$ git commit -m "Initial commit"
[main (root-commit) fda21f5] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 Readme.md

itstw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
$
```

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS COMMENTS
itstw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
$ git log
commit fda21f53b4fb388c35fd0a393ef4792da0d3ab57 (HEAD -> main)
Author: Harshad Prajapati <harshad.b.prajapati@gmail.com>
Date: Sat Jun 1 18:12:37 2024 +0530

    Initial commit

itstw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
$
```

6. Create and Work on a Feature Branch

```
git checkout -b feature/jeera-rice
```

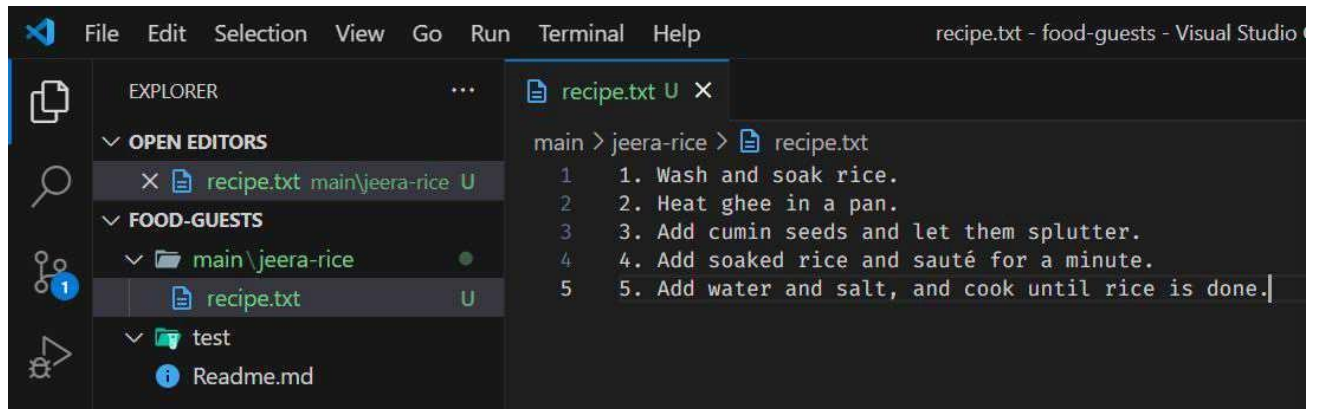
```
# (Create and edit recipe files)
```

```
git add .
```

```
git commit -m "feat(jeera-rice): add steps of preparing Jeera Rice"
```

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS COMMENTS
itstw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
$ git checkout -b feature/jeera-rice
Switched to a new branch 'feature/jeera-rice'

itstw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (feature/jeera-rice)
$
```



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS  COMMENTS

itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (feature/jeera-rice)
$ git commit -m "feat(jeera-rice): add steps of preparing Jeera Rice"
[feature/jeera-rice 8b502d8] feat(jeera-rice): add steps of preparing Jeera Rice
1 file changed, 5 insertions(+)
create mode 100644 main/jeera-rice/recipe.txt

itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (feature/jeera-rice)
$
```

Connecting and Pushing to a Remote GitHub Repository

1. Configure SSH for GitHub

(In your .ssh directory)

```
ssh-keygen -t ed25519 -C "email@gmail.com"
```

(Copy the public key content)

```
MINGW64:/c/Users/itw2a/.ssh

itw2a@DESKTOP-TR1890T MINGW64 ~/.ssh
$ ssh-keygen -t ed25519 -C "harshad.b.prajapati@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/itw2a/.ssh/id_ed25519): |
```

```
MINGW64:/c/Users/itw2a/.ssh

$ ssh-keygen -t ed25519 -C "harshad.b.prajapati@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/itw2a/.ssh/id_ed25519): id_ed25519_drhbp
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in id_ed25519_drhbp
Your public key has been saved in id_ed25519_drhbp.pub
The key fingerprint is:
SHA256:Wh0j0C/OGcJT21eY4KUX9I78gn2b+1ok3mYHZUC8wEU harshad.b.prajapati@gmail.com
The key's randomart image is:
+---[ED25519 256]---+
|      .  .  =E      |
|      .  .  .O      |
|      + O = ..      |
|      O X * +       |
|      X B S         |
|      . B B = .      |
|      * B @ .        |
|      = @ +          |
|      *++           |
+-----[SHA256]-----+

itw2a@DESKTOP-TR1890T MINGW64 ~/.ssh
$
```

```
MINGW64:/c/Users/itsw2a/.ssh
itsw2a@DESKTOP-TR1890T MINGW64 ~/.ssh
$ ls
id_ed25519 id_ed25519.pub id_ed25519_drhbp id_ed25519_drhbp.pub
itsw2a@DESKTOP-TR1890T MINGW64 ~/.ssh
$ |
```

2. Create a New Repository on GitHub

Go to your GitHub account and create a new, empty repository. Do **not** initialize it with a README or .gitignore file.

3. link the Local and Remote Repositories

git remote add origin git@github.com:<username>/food-guests.git

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS COMMENTS
itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
• $ git remote add origin git@github.com:drhbp:harshadbprajapati/food-guests.git
itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
○ $
```

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS COMMENTS
itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
• $ git remote -v
origin git@github.com:drhbp:harshadbprajapati/food-guests.git (fetch)
origin git@github.com:drhbp:harshadbprajapati/food-guests.git (push)
itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
○ $
```

4. Push Branches to GitHub

git push -u origin main

git push --all origin

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS COMMENTS

itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
$ git push origin -u main
Enter passphrase for key '/c/Users/itsw2a/.ssh/id_ed25519_drhbp':
Enumerating objects: 28, done.
Counting objects: 100% (28/28), done.
Delta compression using up to 16 threads
Compressing objects: 100% (17/17), done.
Writing objects: 100% (28/28), 6.37 KiB | 815.00 KiB/s, done.
Total 28 (delta 4), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (4/4), done.
To github.com:drhbp:harshadbprajapati/food-guests.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

itsw2a@DESKTOP-TR1890T MINGW64 ~/projects/devops/food-guests (main)
$
```

